# Section 2.3: Addition and Subtraction of Matrices

Finite Mathematics MTH 132 Spring 2014

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# Matrix Dimensions

$$\begin{bmatrix} 1 & 0 & -3 \\ 4 & -2 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & -3 \\ 4 & -2 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 0 & -3 \\ 4 & -2 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 \\ 2 \\ 4 \\ 0 \end{bmatrix}$$

# Adding and Subtracting

$$\begin{bmatrix} 1 & 4 & 0 \\ 0 & -1 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 2 \\ -2 & -1 & 2 \end{bmatrix}$$

# Adding and Subtracting

$$\begin{bmatrix} 1 & 4 & 0 \\ 0 & -1 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 2 \\ -2 & -1 & 2 \end{bmatrix}$$

# Adding and Subtracting

$$\begin{bmatrix} 1 & 4 & 0 \\ 0 & -1 & 3 \end{bmatrix} + \begin{bmatrix} 2 & -2 \\ 0 & -1 \\ 2 & 2 \end{bmatrix}$$

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b. At the end of 2 weeks, their lengths were 10.2, 11.4, and 11.4 and weights were 196, 196, and 225 respectively. Find a matrix representing how much the rats grew.

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**b.** At the end of 2 weeks, their lengths were 10.2, 11.4, and 11.4 and weights were 196, 196, and 225 respectively. Find a matrix representing how much the rats grew.

$$\begin{bmatrix} 5.6 & 6.4 & 6 \\ 144 & 138 & 149 \end{bmatrix}$$